

0 1 . 1

Describe the induced-fit model of enzyme action **and** how an enzyme acts as a catalyst.

[3 marks]

0 1 . 2

Scientists investigated the action of the enzyme ATP synthase. They made reaction mixtures each containing:

- ATP synthase
- buffer (to control pH)
- substrates.

One of the substrates required in these reaction mixtures is inorganic phosphate (Pi).

Tick (✓) **one** box to show which other substrate the scientists must add to the reaction mixtures to produce ATP.

[1 mark]

Adenine

☐

Adenosine diphosphate

☐

Glucose

☐

Ribose

☐

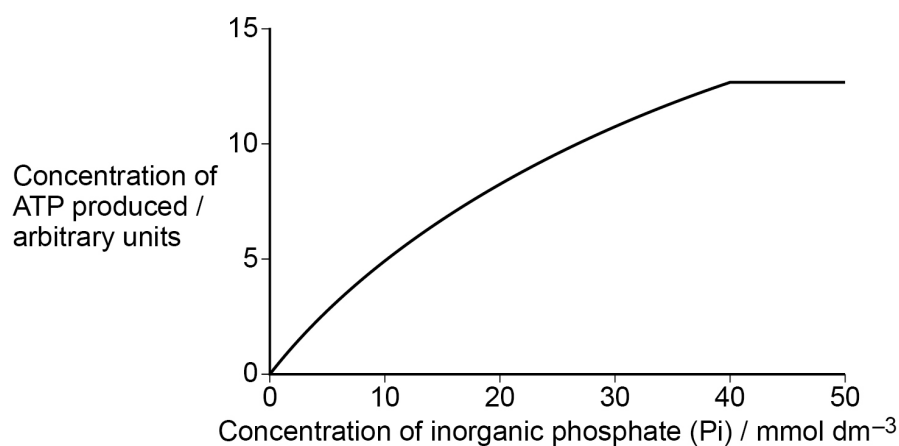
0 1 . 3

The scientists investigated the effect of concentration of inorganic phosphate (Pi) on ATP synthase activity.

After 2 minutes, they stopped each reaction and then measured the concentration of ATP.

Figure 1 shows the scientists' results.

Figure 1



Suggest and explain a procedure the scientists could have used to stop each reaction.

[2 marks]

0 1 . 4

Explain the change in ATP concentration with increasing inorganic phosphate concentration.

[2 marks]
